Ziji Chen

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Education

University of Oxford — Oxford, UK
DPhil in Engineering Science — *Oct 2023 – Expected late 2027/early 2028*
DPhil Affiliate, Hardware AI Governance Lab (HAIGL), Oxford Martin AI Governance Initiative

- Thesis: Understanding, Diagnosing, and Predicting ML System Behaviour. Supervisor: Prof. Noa Zilberman.
- ARIA Research Studentship (2025–2026) and Recognition Award (2026); UKRI supercomputing allocation (2024).

Imperial College London — London, UK
MSc in Applied Computational Science and Engineering — *Oct 2022 – Oct 2023*

Sun Yat-sen University — Guangzhou, China
BSc in Atmospheric Sciences (Minor in Accounting) — *Sep 2017 – Jul 2021*

- GPA 3.7/4.0; National Encouragement Scholarship (2019–20); university scholarships (2017–18, 2019–20).
- National 2nd Prize, CUMCM (China's national modeling contest); Honorable Mention, MCM/ICM (COMAP); National 3rd Prize, Share Cup (national S&T resource-sharing innovation contest).

Research Projects & Publications

Seeing is believing: The case for hardware-centric inefficiency detection and remediation
ACM Symposium on Cloud Computing (SoCC 2026) — *First author*
Keywords: ML observability; anomaly detection; automated remediation

- Built an operator-side system that **detects, attributes and fixes silent ML slowdowns**, such as GPU power throttling that reduces throughput while jobs keep running, using **host-visible telemetry** (e.g. `perf`, `/proc`, `nvidia-smi`...); no guest instrumentation or dedicated healthy-baseline run required.
- Profiled 700+ host metrics across 30 NLP/LLM/CV workloads; identified a **compact, representative profiling set** of about 400 metrics per host. Collection adds about 1.3% CPU and **under 2% end-to-end overhead**.
- Developed a setup-agnostic **online anomaly-detection ensemble** with **fingerprint matching** against a knowledge base of known inefficiencies (*GPU power throttling, huge-page loss, NUMA and IRQ misplacement*).
- On LLaMA-3.1-70B and Qwen3-235B-A22B serving, detected GPU throttling and memory fragmentation in **about 30 s** and fixed them in place, **recovering a 7% throughput loss** and **cutting latency by up to 43%**.

AIGENT: Non-Invasive Classification of ML Workloads from Host Telemetry
Under review, Machine Learning: Science and Technology — *First author*
Keywords: Workload classification; time-series modelling; technical AI governance

- Built a classifier for infrastructure providers that infers *workload type*, *phase*, *task*, *model* and *model–task* category from host telemetry collected **outside VMs and containers**, enabling providers to **verify governance-relevant workload properties** without accessing tenant code, data or model weights.
- **Extra Trees permutation-importance selection** cut about 150 host metrics to 10; **Conformer/BiLSTM** classifiers on 0.5 s windows reach **99.86% accuracy** for LLM vs. non-LLM, over 98% for phase and task, and over 93% across 14 models and 22 model–task combinations, with **under 1 ms inference** on a single CPU core.
- Classification-aware scheduling on an HPC testbed reduced node-hours by up to 3.59% versus random placement.

BluePrism: Application-Aware Isolation and Scheduling on BlueField-3 DPUs

Under review, EuroSys 2027 *Second author (First student author)*

Keywords: DPU isolation; queue scheduling; network offload

- Built a SmartNIC runtime (11.2K LoC, **NVIDIA DOCA**) that decouples hardware-backed isolation (**Scalable Functions**) from per-queue scheduling (**DRR/WDRR**), steering flows with **DOCA Flow and offloaded OVS rules** so application priorities change at runtime without rebuilding protection domains.
- Near line rate on 100 Gbps ports (**99 / 92 Gbps** in host / DPU mode at 1400 B) and up to **65,280 RX/TX queue pairs**; queue remapping completes in **under 2 ms at P99**.
- Under contention, raised video processing from **2 to 25 FPS**, cut priority LiDAR latency from **315 to 119 ms**, kept all 256 federated-learning clients in all 13 rounds, and quarantined anomalous flows in **10.3 ms** (median).

Experience

Research Assistant, Networked Systems (GLASS Project) Nov 2024 – Present

University of Oxford *Oxford, UK*

Keywords: LLM systems; CPU–GPU orchestration; trace analysis

- Studying **host-side orchestration bottlenecks** in accelerator-dense clusters, where accelerators scale faster than the CPUs coordinating them and **MoE/agentic serving** adds per-step host control.
- Identified **control amplification** in DeepSeek-V3/R1 inference traces (H800, EP=32/128): GPUs are over 93% busy, yet prefill issues **181× more control events** than communication and decode has a heavy per-step control tail (P99 0.61 ms); about **60% of control time** re-derives per-phase fixed structure.
- Collected **hardware telemetry** and **Nsight Systems (`nsys`) traces** from widely used ML training and inference workloads to characterise CPU/GPU utilisation, control-event patterns and their impact on scaling.
- These findings motivated the group's **phase-level orchestration design** for LLM inference, which moves repeated host control onto the **fabric and accelerators**.
- Built an **emulation framework** scaling to **1,000 nodes** to evaluate architectures and orchestration; turned ML workload and communication patterns into requirements for architecture, networking and photonics teams.

Linux Cloud Computing & Virtualisation Engineer Jul 2021 – Aug 2022

Tencent Technology *Shenzhen, China*

Keywords: Linux virtualisation; production diagnostics; diagnostic automation

- On the team responsible for **Linux virtualisation** across all of Tencent Cloud (27 regions, 66 availability zones in 2021), resolving **production incidents** in CPU/GPU, memory, I/O and networking.
- Built **diagnostic tools** for recurring failure classes, raising tickets resolved per engineer by **40%**.

Technical Skills

Languages & Systems: Python, C/C++, Bash; Linux internals, virtualisation, Git, CI/CD.
ML Training & Serving: PyTorch, Hugging Face Transformers/PEFT, DDP/FSDP, LoRA, mixed precision; vLLM, MoE/expert parallelism, CUDA graphs, KV-cache management, continuous batching.
Profiling: Nsight Systems/Compute, PyTorch Profiler, `perf`, eBPF, DCGM, `nvidia-smi`, Prometheus/Grafana.
Networking: NCCL, MPI; PCIe/NVLink, RDMA, InfiniBand/RoCE; BlueField-3, DOCA, DPDK, SR-IOV.
Infrastructure: Slurm, Ray, Docker, Kubernetes; Tencent Cloud.

Service & Teaching

Shadow Program Committee Member, ACM EuroSys 2025	Dec 2024 – Feb 2025
Panel Organiser, OxAI 2nd Mini-Conference	Jan – Mar 2025
Python Coding Workshop Instructor, Oxford Summer School	Aug 2024
Teaching Assistant, Programmable Networks, University of Oxford	May 2024
Master's Thesis Mentor, University of Oxford	Oct 2023 – May 2024